Project Design Phase
Proposed Solution Template

Date	12 October 2026
Team ID	SWUID2026403463
Project Name	HotelPro - Online Hotel Room Booking & Reservation Management System
Maximum Marks	2 Marks

Proposed Solution for HotelPro

S. No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Travellers struggle to compare hotels, check live room availability and get quick booking confirmations in one place, while phone calls, front-desk bookings and scattered travel apps cause delays, double bookings, unclear cancellations and delayed refunds. Hotels lack simple tools to manage rooms, rates and live occupancy, which leads to empty rooms and lost revenue.
2	**Idea / Solution Description**	HotelPro is a full-stack web application (React frontend + Spring Boot REST backend + PostgreSQL/MySQL) where guests search by destination and dates, choose a hotel and room type, view availability on a room calendar, add extras, apply coupons, pay through a mock gateway and receive a QR booking voucher. Rooms are held for 10 minutes during checkout so two guests can never book the same room for the same dates. Hotel managers and admins manage rooms, rates, offers, refunds and analytics through role-based dashboards.
3	**Novelty / Uniqueness**	- Real-time room availability with atomic inventory locking and automatic release on timeout - Single platform for room booking, add-on services, dining and spa and event-hall reservations - Policy-based cancellation (up to 24 hours before check-in) with refund preview and tracking - Signed QR vouchers that front-desk staff can verify at check-in and mark as used - Basic recommendation engine using bookings, destinations, room types and ratings - Live occupancy and revenue analytics for hotel managers

S. No.	Parameter	Description
4	**Social Impact / Customer Satisfaction**	- Saves time by removing phone calls and front-desk queues and enabling booking from any device - Fair room allocation and transparent pricing, taxes, fees and refunds - Gives local and independent hotels affordable digital reach - Reviews and ratings help guests choose with confidence - Timely email and in-app reminders reduce missed check-ins
5	**Business Model (Revenue Model)**	- Convenience fee on every booking made - Commission from hotels on room bookings - Paid featured listings and promotional banners - Sponsored offers and coupons from brands and travel partners - Premium hotel plans with advanced analytics
6	**Scalability of the Solution**	- Stateless REST backend that scales horizontally behind a load balancer, with Redis for shared room locks and caching - Designed to grow to 100,000 users, 500 hotels and 50,000 bookings per day with indexing and pagination - Modular architecture for adding real payment gateways, SMS and push notifications - Future mobile apps, dynamic pricing, loyalty programs and multi-language support